

Section A

1. (a) What is the basic principle of SDMA?

Ans: SDMA (Space Division Multiple Access) allows multiple users to communicate by using different physical locations or separate spatial channels.

(b) Which multiple-access technique assigns different frequency bands to users?

Ans: FDMA (Frequency Division Multiple Access).

(c) In TDMA, how is the channel divided among multiple users?

Ans: The channel is divided into time slots, and each user is assigned a specific time slot.

(d) What type of codes are used to ~~diff~~ distinguish users in CDMA?

Ans: Unique spreading (PN) codes are used.

(e) A signal is intentionally spread over a wider bandwidth to reduce interference. Name the technique.

Ans: Spread spectrum.

(f) A modulation scheme varies carrier amplitude according to the message signal. Name it.

Ans: Amplitude Modulation (AM).

(g) Several signals must share one communication medium without mutual interference. What process combines them?

Ans: Multiplexing

- Q. A wireless service area is divided into small geographic regions to enable frequency reuse. What are these regions called?

Ans. Cells.

Section B

- Q. Analyze a GSM call path and state the roles of BTS, BSC and MSC.

Ans. A GSM call starts from the mobile phone and reaches the BTS (Base Transceiver Station), which provides the wireless connection. The BSC (Base Station Controller) manages multiple BTSs, allocates radio resources, and controls handovers. The MSC (Mobile Switching Center) establishes, routes, and manages calls between users and other networks.

- Q. Analyze how GSM handover and authentication support continuity and security.

Ans. Handover transfers an active call from one cell to another without disconnecting it, ensuring continuous communication while moving. Authentication verifies the subscriber using the SIM card and secret keys, preventing unauthorized access and ensuring secure communication.

3. (a) Analyze the GPRS architecture and state the function of SGSN and GGSN.

Ans GPRS is a packet-switched data service used in GSM networks. The SGSN tracks the user's location, authenticates users, and delivers data packets. The GGSN connects the GPRS network to external networks such as the Internet and routes data packets between mobile users and the Internet.

(b) Analyze how localization / calling and mobility support help a subscriber maintain service while moving.

Ans Localization keeps track of the subscriber's current location so that calls and messages can be delivered correctly. Mobility support updates the subscriber's location and performs handovers between cells, allowing uninterrupted communication while moving from one coverage area to another.